

Risk Optimization - Facility Allocation

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1 Introduction

This project addresses the Facility Location Problem under the complexity of stochastic demand. While the primary objective is to minimize fixed opening and connection costs, the central challenge lies in defining feasibility when demand is uncertain. We investigate frameworks to quantify reliability and control the risk of unmet demand. Specifically, we evaluate the trade-offs involved in moving from a conservative worst-case approach to more flexible probabilistic chance constraints and loss-based risk measures.

1.1 Data Generation

We distribute $|\mathcal{J}|$ independent demand points, ξ_j , uniformly in $[0, 100]^2$. Uncertain demands are modeled as truncated normal distribution with $\mu_j \sim U[10, 50]$ and coefficients of variation $\sigma_j/\mu_j \sim U[0.05, 0.35]$. Facility locations $\mathcal{I}$ are generated so the first three are placed within a 10×10 neighborhood of the demand nodes with the highest μ_j , while the remainder are distributed uniformly in a 60×60 region centered at $(50, 50)$. Finally, we define fixed costs $F_i = O|\mathcal{J}|$ and variable costs $C_{ij} = D\|\text{loc}_i - \text{loc}_j\|_2$.

2 Worst-Case Scenario Facility Location

2.1 Introduction and Methodology

We first address the uncertain demand ξ using a worst-case optimization approach. As outlined in the lecture notes, this involves identifying a feasible set, provided one exists, where constraints are satisfied under all possible outcomes, effectively aiming for zero tolerance regarding constraint violations.

We approximate the continuous distribution of demand using a sample set $\{\xi^1, \dots, \xi^K\}$ of independent realizations, a method known as Sample Average Approximation or SAA in the lecture notes. This allows us to reformulate the problem as a Mixed-Integer Linear Program.

2.2 Mathematical Formulation

Let $x_i \in \{0, 1\}$ be the binary variable indicating whether facility $i \in \mathcal{I}$ is open, and let $y_{ij} \in \{0, 1\}$ indicate whether demand point $j \in \mathcal{J}$ is assigned to facility i . The parameters include fixed opening costs F_i , connection costs C_{ij} , and facility capacities $M_i = M|\mathcal{J}|$ for a fixed constant M . The objective is to minimize total fixed and connection costs and serve demand.

The worst-case scenario MILP is formulated as follows

$$\min_{x,y} \sum_{i \in \mathcal{I}} F_i x_i + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} C_{ij} y_{ij} \quad (1)$$

Subject to

$$\sum_{i \in \mathcal{I}} y_{ij} = 1 \quad \forall j \in \mathcal{J} \quad (2)$$

$$y_{ij} \leq x_i \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{J} \quad (3)$$

$$\sum_{j \in \mathcal{J}} \xi_j^k y_{ij} \leq M_i x_i \quad \forall i \in \mathcal{I}, \forall k \in \{1, \dots, K\} \quad (4)$$

$$x_i \in \{0, 1\}, \quad y_{ij} \in \{0, 1\} \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{J} \quad (5)$$

Constraint (4) embodies the Worst-Case Analysis approach described in the theoretical framework. A worst-case formulation seeks to minimize the objective subject to constraints holding for the essential supremum of the random variable.

Since we approximate the uncertainty using a discrete support of K scenarios, the essential supremum reduces to the maximum over the observed set. By enforcing the inequality for all $k \in \{1, \dots, K\}$, we effectively require

$$\max_{k \in \{1, \dots, K\}} \left(\sum_{j \in \mathcal{J}} \xi_j^k y_{ij} - M_i x_i \right) \leq 0 \quad \forall i \in \mathcal{I}$$

This ensures that the chosen capacity is robust against the most adverse realization of demand within our sample. Consistent with the lecture notes, this approach is highly conservative. It aims to eliminate the risk of capacity violation entirely, regardless of how unlikely the worst-case scenario might be.

2.2.1 Numerical Analysis

We tested configurations varying M (Table 1) with $\mathcal{J} = 50$ and $\mathcal{I} = 10$, using the same parameters for opening cost, $O = 500$ and $D = 270$ for link cost throughout the analysis for consistency. For feasible configurations, increasing M allows the solver to consolidate demand into fewer facilities, reducing fixed costs but increasing connection costs as average distance to facilities grows. We let facility capacity be equal to $|\mathcal{I}_{open}| \cdot (M \cdot |\mathcal{J}|)$. The model is implemented in Python using gurobipy with $K = 1000$ realizations.

Table 1: Worst case Optimization Results (varying M , fixed $K = 1000$)

M	Open	Fixed Cost	Conn Cost	Total Cost	Max Dem	Avg Dem	Facility Cap
4.0	10	250,000.0	229,008.8	479,008.8	1,595.2	1,442.1	2,000.0
6.0	7	175,000.0	223,076.8	398,076.8	1,595.2	1,442.1	2,100.0
8.0	5	125,000.0	245,121.6	370,121.6	1,595.2	1,442.1	2,000.0

The worst case approach is highly conservative. For $M = 4.0$, the cheapest, robust solution is to open all 10 facilities, with a total supply of 2000, to cover a worst-case

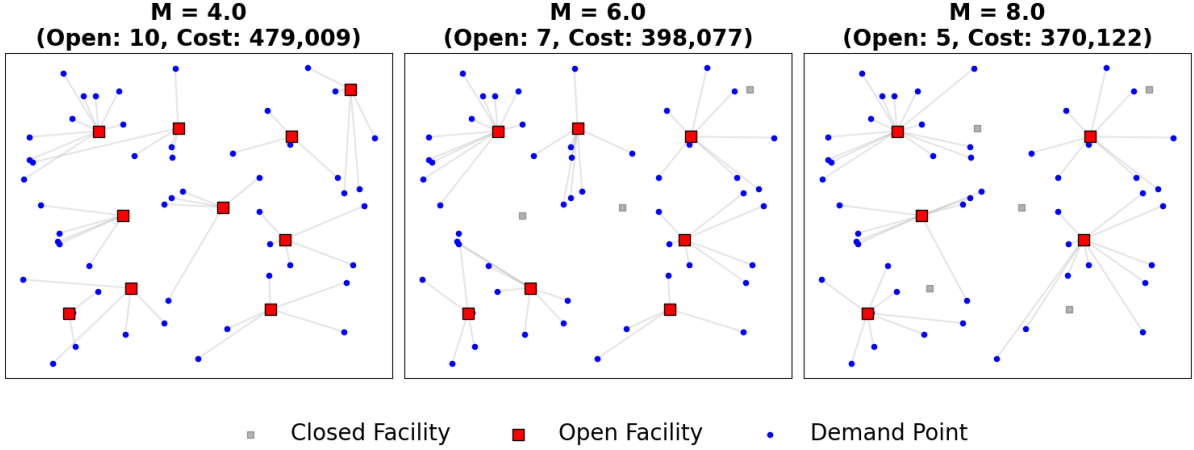


Figure 1: Optimal topology for $M \in \{4, 6, 8\}$. We notice as capacity, M increases, less facilities are opened and distances increase, making connection cost rise.

demand of 1,595.2, with avg demand of about 1,442.1. This results in low utilization rates during normal scenarios.

2.3 Extreme Worst Case Events

In the previous analysis, we relied on the assumption that the demands ξ_j associated with each demand point $j \in \mathcal{J}$ are independent random variables. This assumption has profound implications for the probability of extreme systemic events.

Mathematically, independence implies that the joint probability of all demands simultaneously exceeding a critical threshold x is the product of their marginal probabilities:

$$P\left(\bigcap_{j=1}^{|\mathcal{J}|} (\xi_j > x)\right) = \prod_{j=1}^{|\mathcal{J}|} P(\xi_j > x) \quad (6)$$

Since typically $P(\xi_j > x) < 1$ for large x , this product decays fast with the number of demand points $|\mathcal{J}|$. Under this probabilistic view, planning for a simultaneous system-wide peak appears excessively conservative, as it is an event that, statistically, never happens.

2.3.1 The Failure of Independence

In many practical applications, however, this independence assumption is dangerously flawed. Consider an energy grid capacity planning problem in Southern Europe during a severe heatwave. In such scenarios, the electricity demands ξ_j are heavily positively correlated.

If one region experiences a spike in consumption due to air conditioning (ξ_1 is high), it is highly probable that neighboring regions ($\xi_2, \dots, \xi_j$) are experiencing the same surge. In the extreme case of perfect positive correlation, the joint probability does not decay, rather, for large load x it might be:

$$P\left(\bigcap_{j=1}^{|\mathcal{J}|}(\xi_j > x)\right) \approx P(\xi_1 > x) \quad (7)$$

While Sample Average Approximation (SAA) inherently captures the dependence structure of the underlying distribution, relying on a finite sample size K creates a critical vulnerability regarding tail dependence. Systemic shocks, where highly correlated demands peak simultaneously, are often rare events located far in the joint tail of the distribution. In a limited sample (e.g., $K = 100$), it is statistically unlikely that these specific joint realizations will be observed.¹

2.3.2 Theoretical Justification via Minkowski's Inequality

To immunize the solution against correlation uncertainty and systemic shocks for finite K , we adopt an even more conservative formulation. Rather than optimizing against specific scenarios where demands might offset one another, we require the system to accommodate the maximum possible (observed) demand for each ξ_j individually.

We justify this approach mathematically using the Minkowski Inequality for the L_∞ -norm. Let the worst-case magnitude of the random demand $\boldsymbol{\xi}_j = [\xi_j^1 \ \xi_j^2 \ \dots \ \xi_j^K]^\top$, be defined as:

$$\|\boldsymbol{\xi}_j\|_\infty = \max_{k \in \{1, \dots, K\}} |\xi_j^k| = \max_{k \in \{1, \dots, K\}} \xi_j^k$$

Notice that, in our case, this expression reduces to the maximum and coincides with the maximum demand in the same sense as in Section 2.2, since demands are non-negative. Our objective is to ensure that the capacity M_i of a facility is sufficient to handle the aggregate demand of all assigned customers.

If y_{ij} represents the assignment of demand, the total load on facility i is $D_i = \sum_{j \in \mathcal{J}} y_{ij} \boldsymbol{\xi}_j$. The actual worst-case load, the maximum total demand the facility ever faces, is given by the uniform norm of the sum:

$$\left\| \sum_{j \in \mathcal{J}} y_{ij} \boldsymbol{\xi}_j \right\|_\infty \quad (8)$$

By the Minkowski inequality, which holds for $p \geq 1$ (including $p = \infty$), the norm of a sum is bounded by the sum of the norms:

$$\left\| \sum_{j \in \mathcal{J}} y_{ij} \boldsymbol{\xi}_j \right\|_\infty \leq \sum_{j \in \mathcal{J}} y_{ij} \|\boldsymbol{\xi}_j\|_\infty \quad (9)$$

This inequality highlights the fundamental difference between the standard scenario-based approach and this conservative formulation. The left-hand side represents the true worst-

¹I got the inspiration for this from an actuarial textbook I used in my second year. In particular, a mathematical discussion of heavy-tailed distributions and why standard arguments can fail under extreme dependence can be found in Thomas Mikosch, *Non-Life Insurance Mathematics: An Introduction with Stochastic Processes* (Springer, 2004), Chapter 3, Section 3.2.

case realization, which is sensitive to the correlation structure; if demands are independent or negatively correlated, peaks in ξ_j may not occur simultaneously, reducing the total load. The right-hand side represents the sum of individual worst-case peaks, where all demands maximize simultaneously. By enforcing feasibility for the right-hand side, we establish a conservative upper bound on the risk. This guarantees that the capacity constraints are satisfied regardless of the underlying dependency structure.

2.4 Computational Efficiency and Asymptotic Stability

A distinct practical advantage of this component-wise formulation over the standard formulation is its scalability. By decoupling the sample size from the optimization constraints, the method permits a rigorous analysis of the system's behavior under extreme sample sizes.

In a standard model, the optimization solver must explicitly enforce capacity constraints for every scenario $k \in \{1, \dots, K\}$ and for every facility $i \in \mathcal{I}$. For large sample sizes necessary to capture tail events, the resulting Mixed-Integer Linear Program (MILP) becomes computationally intractable due to the explosion in the number of constraints.

In contrast, the component-wise approach shifts the computational burden from the solver to a pre-processing step. We pre-calculate the scalar worst-case value $\xi_j^{\max} = \max_k \xi_j^k$ for each demand point. The MILP then solves a deterministic problem using these aggregates. Consequently, the number of constraints depends only on the number of facilities $|\mathcal{I}|$, remaining constant even as K scales to billions. This reduction allows us to bypass the computational limitations inherent in scenario-based optimization.

Leveraging this efficiency, we analyzed the stability of the total system demand, defined as $D_{sys}(K) = \sum_j \max_{k \leq K} \xi_j^k$, as K increases from 10 to 1,000,000,000.

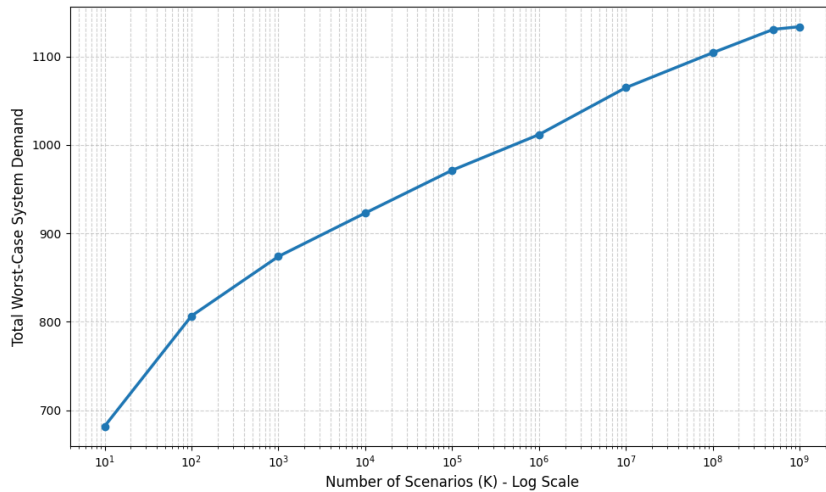


Figure 2: Asymptotic stability of Total Worst-Case System Demand $D_{sys}(K)$ with $|\mathcal{J}| = 20 \neq 50$ for computational purposes. The logarithmic scale on the x-axis highlights that as the sample size K approaches 10^9 , the marginal growth in worst-case demand diminishes.

While the aggregated worst-case demand rises sharply for small K , the rate of growth diminishes significantly as K increases (although it still diverges). Thus, despite being a conservative approximation, the solution levels to a somewhat stable value rather than

inflating costs arbitrarily with additional data. It is important to note that the standard "Max of Sum" formulation (8) also theoretically tends toward infinity as K does, though at a much slower rate. This behavior is consistent with the properties of SAA. As the sample size K increases, the optimal value of the SAA problem converges to the optimal value of the true problem. Since the underlying demand follows a Normal distribution truncated at zero, the support is unbounded towards positive values. Therefore, as $K \rightarrow \infty$, the sampling of increasingly large values (outliers, in some sense) is guaranteed, and the true worst-case scenario converges to infinity (the essential supremum). This implies that the formulation must eventually become infeasible (diverge) as demand exceeds fixed capacities.

It is particularly noteworthy that for $K=100$, the total worst-case demand calculated via individual maxima exceeds 800. Given that the maximum available system capacity for $M=4$ is capped at $20(4 \cdot 10) = 800$, this configuration effectively becomes infeasible. This was not the case following a simulation with $(K, I, J) = (100, 10, 20)$ with the usual constraint, 4, however upping $K = 1000$ confirms it is indeed infeasible like the above shows at less K .

3 Chance-Constrained Facility Location

3.1 Methodology

The worst-case approach ensures absolute robustness but may lead to excessive costs by effectively optimizing for outliers. In practice, as discussed in Chapter 3 of the lecture notes, it is often preferable to enforce a Probabilistic Constraint, also known as a Chance Constraint. We require that the probability of satisfying all capacity constraints simultaneously is at least α .

We formulate this as a Joint Chance Constraint rather than a set of individual chance constraints. As noted in Example 3.2 of the lecture notes, satisfying individual constraints under facility-specific scenarios K does not guarantee jointly system-level reliability. Our requirement is

$$P \left(\sum_{j \in \mathcal{J}} \xi_j y_{ij} \leq M_i x_i, \forall i \in \mathcal{I} \right) \geq \alpha$$

Following the methodology for discrete distributions outlined in Section 3.1.1, we reformulate this joint probabilistic constraint into a Mixed-Integer Linear Program.

3.2 Mathematical Formulation

We introduce a binary variable $z_k \in \{0, 1\}$ for each scenario $k \in \{1, \dots, K\}$ to indicate compliance. Following the convention in Section 3.1.1 of the lecture notes, we define $z_k = 1$ to imply that the system satisfies the capacity constraints for all facilities in scenario k .

The MILP formulation for the joint chance constraint is:

$$\min_{x,y,z} \sum_{i \in \mathcal{I}} F_i x_i + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} C_{ij} y_{ij} \quad (10)$$

Subject to the network constraints (1.1b) and (1.1c):

$$\sum_{i \in \mathcal{I}} y_{ij} = 1 \quad \forall j \in \mathcal{J} \quad (11)$$

$$y_{ij} \leq x_i \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{J} \quad (12)$$

And the probabilistic constraints:

$$\sum_{j \in \mathcal{J}} \xi_j^k y_{ij} \leq M_i x_i + M_k^{big} (1 - z_k) \quad \forall i \in \mathcal{I}, \forall k \in \{1, \dots, K\} \quad (13)$$

$$\sum_{k=1}^K z_k \geq \lceil K\alpha \rceil \quad (14)$$

$$x_i, y_{ij}, z_k \in \{0, 1\} \quad (15)$$

Constraint (13) uses the method from section 3.1.1. To ensure numerical stability and tightness, we calculate M_k^{big} based on the maximum possible violation in scenario k , as suggested in the lecture notes (eq 1.12). Since the demand assigned to a facility cannot exceed the total system demand, we define:

$$M_k^{big} = \sum_{j \in \mathcal{J}} \xi_j^k$$

Constraint (14) is the integer programming equivalent of the probability requirement $\frac{1}{K} \sum_k z_k \geq \alpha$, ensuring that at least $\lceil K\alpha \rceil$ scenarios are fully satisfied.

3.3 Analysis of Results

We analyze the impact of the reliability parameter α on the solution cost. We compare the Robust solution (implicitly $\alpha = 1.0$) with relaxed requirements $\alpha \in \{0.95, 0.90\}$. The results are summarized in Table 2.

Table 2: Chance-Constrained Results (fixed $M = 4.0$, varying α)

α (Reliability)	Open Fac.	Satisfied Scenarios	Total Cost	Reduct. (%) from $\alpha = 1$
1.00 (Worst case)	10	1000	479,008.8	-
0.95	9	950	450,687.45	6.12%
0.90	9	900	444,482.29	8.31%

Reducing the requirement to satisfying 95% of scenarios allows the model to ignore the most extreme demand spikes. This demonstrates the price of robustness discussed in the lectures where eliminating the last few percent of risk requires disproportionately high investment.

4 CVaR-Constrained Facility Location

4.1 Loss Function Formulation

In the previous section, we addressed the risk of capacity violation by constraining the probability of occurrence (Chance Constraints). However, as noted in the introduction of the lecture notes, probabilistic constraints focusing solely on frequency ($P(\text{failure}) \leq \alpha$) ignore the magnitude of the consequences in the tail distributions. In a cost minimization context, with $\alpha = 0.95$, this leads the model to strategically discard the most expensive 5% of scenarios, effectively ignoring the constraints where demand is highest, without providing any metric on the severity of the resulting unmet demand.

To address Task c), we must explicitly formulate a loss function $L(x, y, \xi)$ that quantifies the "consequence" of unmet demand. If the demand allocated to facility i exceeds its capacity M_i , the system incurs a loss equivalent to the shortage.

We define the loss in scenario k , denoted $\mathcal{L}_k$, as the total unserved demand across the network:

$$\mathcal{L}(y, x, \xi^k) = \sum_{i \in \mathcal{I}} \left(\sum_{j \in \mathcal{J}} \xi_j^k y_{ij} - M_i x_i \right)_+ = \sum_{i \in \mathcal{I}} u_{ik} \quad (16)$$

where $(\cdot)_+ = \max\{0, \cdot\}$ and u_{ik} is a linearization of the max, which we will introduce shortly. This loss function is convex with respect to the decision variables, as it is the sum of convex functions.

4.2 Risk Measure: Conditional Value-at-Risk (CVaR)

To control this loss, we employ the Conditional Value-at-Risk (CVaR). Unlike Value-at-Risk (VaR), which only provides a threshold for the worst-case frequency, CVaR quantifies the expected loss in the worst $(1 - \alpha)$ percent of cases. This aligns with the goal of minimizing the magnitude of tail risk.

Formally, for a loss distribution with a continuous distribution function, CVaR is defined as:

$$CVaR_\alpha(\mathcal{L}) = \mathbb{E}[\mathcal{L} \mid \mathcal{L} \geq VaR_\alpha(\mathcal{L})] \quad (17)$$

We refrain from using this definition directly for reasons of tractability, favoring instead the alternative minimization formulation because it is convex with respect to the decision variables:

$$CVaR_\alpha(\mathcal{L}) = \min_{z \in \mathbb{R}} \left\{ z + \frac{1}{1 - \alpha} \mathbb{E}[(\mathcal{L} - z)_+] \right\} \quad (18)$$

However, we must still employ Sample Average Approximation because the expectation term $\mathbb{E}[(\mathcal{L} - z)_+]$ in this formulation represents a multidimensional integral that is generally hard to compute. Furthermore, for the discrete distributions resulting from SAA, the original conditional expectation definition would be ill-defined due to probability atoms, which confirms the necessity of using the minimization formulation.

4.3 Mathematical Formulation

We modify the original facility location problem to minimize total costs subject to a constraint on the CVaR of the unserved demand. We utilize the constrained formulation (4.10) in lecture notes rather than minimizing the CVaR directly (4.9) to avoid the difficulty of assigning a monetary price to physical unserved demand, allowing us instead to map a clear efficient frontier between operational cost and safety by systematically varying the risk limit λ .

Following the discrete approximation method for optimization found in Section 4.4 of the lecture notes, we introduce:

- $z \in \mathbb{R}$: the auxiliary variable representing the VaR threshold.
- $w_k \in \mathbb{R}_{\geq 0}$: the shortfall variable capturing the excess loss over z in scenario k .
- $u_{ik} \in \mathbb{R}_{\geq 0}$: the specific capacity shortage at facility i in scenario k .

The MILP formulation is as follows:

$$\min_{x,y,z,w,u} \quad \sum_{i \in \mathcal{I}} F_i x_i + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} C_{ij} y_{ij} \quad (19)$$

Subject to the standard network constraints (1.1b-1.1e):

$$\sum_{i \in \mathcal{I}} y_{ij} = 1 \quad \forall j \in \mathcal{J} \quad (20)$$

$$y_{ij} \leq x_i \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{J} \quad (21)$$

Constraints defining the inner part of Loss Function $\mathcal{L}_k$, that is firstly linearize the $\max(0, x)$:

$$u_{ik} \geq \sum_{j \in \mathcal{J}} \xi_j^k y_{ij} - M_i x_i \quad \forall i \in \mathcal{I}, \forall k \in \{1, \dots, K\} \quad (22)$$

$$u_{ik} \geq 0 \quad \forall i \in \mathcal{I}, \forall k \in \{1, \dots, K\} \quad (23)$$

Here, $\sum_{i \in \mathcal{I}} u_{ik}$ implicitly represents the total loss $\mathcal{L}_k$ for scenario k .

Constraints defining the CVaR auxiliary variables $(\mathcal{L} - z)_+$ (see page 76 Lecture Notes):

$$w_k \geq \left(\sum_{i \in \mathcal{I}} u_{ik} \right) - z \quad \forall k \in \{1, \dots, K\} \quad (24)$$

$$w_k \geq 0 \quad \forall k \in \{1, \dots, K\} \quad (25)$$

The Risk Constraint:

$$z + \frac{1}{1 - \alpha} \mathbb{E}[(L(x, \boldsymbol{\xi}) - z)_+] = z + \frac{1}{(1 - \alpha)K} \sum_{k=1}^K w_k \leq \lambda \quad (26)$$

$$x_i, y_{ij} \in \{0, 1\}, \quad z \in \mathbb{R} \quad (27)$$

In this model, constraint (22) calculates the physical unserved demand. The solver optimizes the topology (x, y) such that the tail expectation of this unserved demand (calculated via z and w_k) satisfies the limit λ . This approach ensures that even in the worst-case scenarios, the magnitude of failure is contained, addressing the shortcomings of the pure chance-constrained approach.

4.4 Numerical Implementation

To evaluate the cost-risk trade-off empirically, we investigate the above for varying levels of the risk limit λ . The resulting optimal objective values (Total Cost) are recorded and plotted against their respective risk limits. This generates the "Efficient Frontier" markowitz alike, visualizing the boundary between financial efficiency and operational safety.

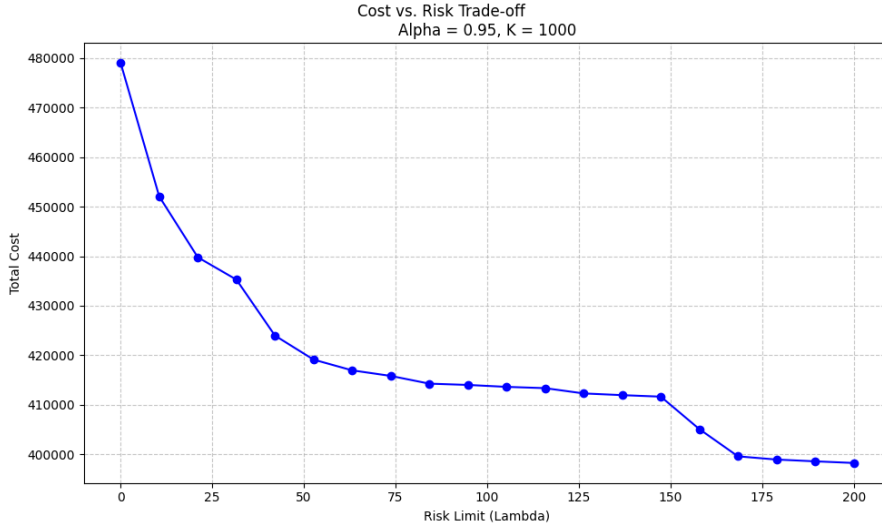


Figure 3: Cost-Risk trade-off curve ($\alpha=0.95$). The steep initial slope indicates that accepting a small amount of tail risk λ (non met demand) allows for a disproportionately large reduction in facility setup costs

By constraining the Conditional Value-at-Risk of this loss, we provide a specific managerial guarantee; for example, setting the risk limit to 75 ensures that the expected average shortage during the worst 5 percent of demand scenarios remains below 75 units. This formulation fundamentally distinguishes the Conditional Value-at-Risk approach from the Chance-Constrained method by strictly controlling the magnitude of potential failures rather than merely limiting their frequency.

At $\lambda = 0$, the model is conservative, opening maximum facilities to eliminate tail risk, which results in the highest total cost. As λ increases, the optimization trades safety for efficiency by consolidating facilities. This lowers fixed costs, but increases exposure to unserved demand, demonstrating the high price of strict robustness.

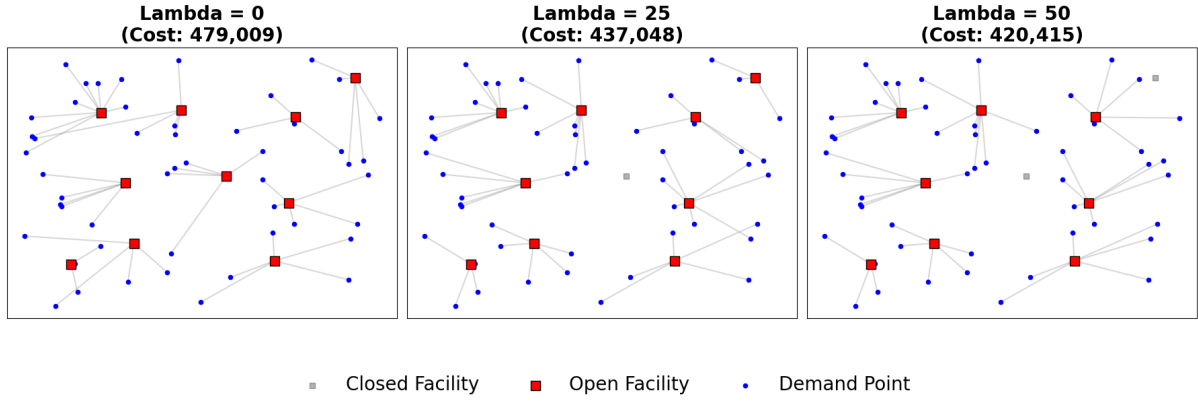


Figure 4: Topology of different $\lambda \in \{0, 25, 50\}$. For $\lambda = 0$, we recover worst-case scenario. Notice how increasing λ closes facilities.

5 Comparative Analysis and Conclusion

The comparative analysis of Worst-Case, Chance-Constrained, and CVaR methodologies reveals that the choice of risk measure serves as a fundamental managerial decision regarding the nature of safety. The Worst-Case approach guarantees absolute feasibility by optimizing against the essential supremum of demand, but this comes at the cost of being prohibitively expensive and highly sensitive to sample outliers. It effectively eliminates risk but ignores the statistical improbability of such extreme simultaneous events.

The Chance-Constrained approach provides a more economically viable alternative by accepting a controlled frequency of failure. However, this method introduces a critical vulnerability, as it is blind to the magnitude of the consequences. By treating a small capacity breach identically to a catastrophic shortage, it fails to distinguish between manageable operational hiccups and systemic failures.

CVaR optimization emerges as the superior framework for rational decision-making. Unlike chance constraints, it explicitly minimizes the expected magnitude of losses in the tail, ensuring that failures, when they occur, are contained. Theoretically, CVaR is a coherent risk measure, meaning it satisfies the axioms of convexity and subadditivity. Practically, this ensures that the model correctly rewards risk diversification and pooling. We therefore recommend the CVaR approach as it offers the only mechanism to balance the probability of failure with the severity of its impact.